

Abstract

This project report presents the manufacturing of a collet using a universal lathe machine as part of the Open Ended Laboratory (OEL) work. A die steel workpiece of initial dimensions 160 mm length and 60 mm diameter was selected. Various machining operations such as turning, taper turning, centring, drilling, boring, and threading were performed to achieve the final dimensions of 132 mm length and 54 mm diameter, with a bore of 2.54 mm and 12 TPI threading. The report discusses the workpiece material, its significance and applications, the reason for selecting a collet, details of the universal lathe machine, machining operations performed, and concludes with the outcomes of the project.

Introduction

In manufacturing and machining processes, precision holding devices play a vital role in ensuring accuracy, repeatability, and safety. One such important work-holding device is the collet, widely used in lathe and milling machines. The objective of this Open Ended Laboratory (OEL) project was to design and manufacture a collet using a universal lathe machine while applying practical machining knowledge.

This project allowed hands on experience with different lathe operations and helped in understanding material behavior, tool selection, dimensional control, and surface finishing. The collet was manufactured from die steel, chosen for its strength and wear resistance, making it suitable for industrial applications.

Manufacturing of a Collet Using Universal Lathe Machine

1. Workpiece Material

1.1 Material Used: Die Steel

The workpiece material selected for this project was die steel, **Die steel** is a special type of alloy steel that is used to manufacture dies, molds, and high-strength machine components. It is designed to withstand high pressure, heavy loads, repeated impact, and wear during manufacturing processes.

In simple words, die steel is a strong, hard, and durable steel used where ordinary steel would wear out or fail, with an initial size of:

- **Length:** 160 mm
- **Diameter:** 60 mm

After machining:

- **Final Length:** 132 mm
- **Final Diameter:** 54 mm

1.2 Significance of Die Steel

Die steel is a high-strength alloy steel known for its:

- High hardness
- Excellent wear resistance
- Good toughness
- Ability to withstand high stress and pressure

These properties make die steel suitable for precision components that experience repeated loading and friction, such as collets.

1.3 Applications of Die Steel

Die steel is commonly used in:

- Dies and molds
- Cutting tools
- Clamping and holding devices
- Automotive and aerospace components
- High-precision machine parts

2. Selection of Collet for the Project

2.1 Why Collet Was Selected

A collet is a precision work-holding device used in machine tools to grip and hold cylindrical or hexagonal workpieces accurately. It is usually fitted into the spindle of a lathe, milling machine, or CNC machine. The collet was selected for this project because:

- It is a precision work holding device
- It requires multiple lathe operations
- It demonstrates practical manufacturing skills
- It is commonly used in real industrial machining setups

Manufacturing a collet helped in understanding tight tolerances, accurate boring, and threading operations.



Figure 1

2.2 Uses of a Collet

A collet is used to:

- Hold round or hexagonal workpieces
- Provide accurate concentric clamping
- Reduce run-out during machining
- Improve surface finish and machining accuracy

2.3 Advantages of Collet

- High gripping accuracy
- Uniform clamping force
- Better concentricity than chucks
- Reduced vibration
- Easy installation and removal

2.4 Applications of Collet

Collets are widely used in:

- Lathe machines
- CNC machines
- Milling machines
- Drilling machines
- Precision manufacturing industries

3. Machine Used: Universal Lathe Machine

3.1 Description of Universal Lathe Machine

A **universal lathe machine** is a versatile machine tool used to perform various machining operations on cylindrical workpieces. It is one of the most commonly used machines in workshops and manufacturing industries.



Figure 2

3.2 Main Parts of a Universal Lathe Machine

- **Bed:** Provides support and rigidity to the machine
- **Headstock:** Holds the spindle and chuck
- **Tailstock:** Supports the workpiece and holds tools like drills
- **Carriage:** Moves the cutting tool along the workpiece
- **Lead Screw:** Used for threading operations
- **Tool Post:** Holds the cutting tool

3.3 Significance of Universal Lathe Machine

- High versatility
- Ability to perform multiple operations
- Suitable for precision machining
- Widely used in tool rooms and workshops

4. Machining Operations Performed

4.1 Facing

Facing is a machining operation performed on a lathe machine in which material is removed from the end face of a rotating workpiece to produce a flat, smooth surface perpendicular to the axis of rotation.

In facing:

- Workpiece rotates
- Cutting tool moves radially (from outside toward the center or vice versa)

Purpose:

- Produce a flat and smooth end surface
- Make the workpiece square to its axis
- Achieve accurate length of the component
- Prepare the workpiece for further operations such as turning, drilling, and threading



Figure 3

4.2 Turning

Turning is a fundamental machining operation performed on a lathe machine in which material is removed from the outer surface of a rotating workpiece using a single-point cutting tool. The cutting tool moves parallel to the axis of rotation to reduce the diameter and produce a smooth cylindrical surface. In this project, turning was performed to reduce the diameter from 60 mm to 54 mm and to achieve the required length.

Purpose of Turning:

- To obtain accurate diameter
- To improve surface finish



Figure 4

4.3 Taper Turning

Taper turning is a machining operation performed on a lathe machine in which the diameter of a rotating workpiece is gradually reduced or increased along its length, producing a conical surface instead of a straight cylindrical one.

In taper turning:

- The **workpiece rotates**
- The **cutting tool moves at an angle** to the axis of the workpiece

Purpose:

- To form tapered surfaces required for collet functionality
- To ensure proper gripping and alignment



Figure 5

4.4 Centring

Centring is a machining operation in which a small conical hole is produced at the center of the workpiece using a centre drill.

It is usually the first internal operation performed before drilling or boring.

Purpose:

- To ensure accurate alignment
- To guide drilling and boring operations
- Locate the **exact center** of the workpiece
- Support the workpiece using tailstock center if required



Figure 6

4.5 Drilling

Drilling is a machining operation used to produce a circular hole in a solid workpiece using a rotating drill bit.

On a lathe:

- Workpiece rotates
- Drill is fed axially using the tailstock

Purpose:

- To initiate the internal hole
- To prepare the workpiece for boring



Figure 7

4.6 Boring

Boring is a machining operation used to enlarge and finish an existing hole using a single point boring tool.

Unlike drilling, boring provides high accuracy and better surface finish.

Purpose:

- To achieve accurate internal diameter
- To improve hole finish and concentricity
- Ensure **concentricity** with outer surface



Figure 8

4.7 Threading

Threading is a machining operation in which helical grooves (threads) are cut on the external or internal surface of a cylindrical workpiece. Threading is performed on a lathe machine using a single-point threading tool or a die/tap. Threads are used to join parts, transmit motion, and provide mechanical locking.

In this project, **12 TPI (threads per inch)** threading was performed.

Purpose:

- To enable assembly and secure fitting
- To provide mechanical locking
- Allow **adjustment and locking** of parts
- Transmit axial force or motion



Figure 9

4.8 CNC Wire Cut

CNC Wire Cut, also known as Wire Electrical Discharge Machining (Wire EDM), is a non conventional machining process in which a thin, electrically charged wire is used to cut intricate shapes in electrically conductive materials.

- The wire acts as an electrode
- The material is removed by sparks (electrical discharges)
- The process is controlled using CNC programming, allowing precise cuts

Unlike conventional machining, Wire EDM does not use mechanical force.

Purpose:

- **Cut slots, profiles, or contours** accurately in the workpiece

- **Work with hard materials** such as die steel, carbide, or tool steel
- **Achieve high dimensional accuracy and excellent surface finish**
- **Avoid mechanical stresses or deformation**, especially for delicate or thin components
- **Reduce manual finishing and improve repeatability**



Figure 10

Finishing Operations

After performing all the primary machining operations on the collet, finishing cuts of 0.2 mm were applied to each feature to achieve higher dimensional accuracy and a smoother surface. These fine cuts help to remove minor imperfections and tool marks left from the roughing operations. In addition to the lathe finishing cuts, the workpiece was carefully polished using sandpaper, which further improved the surface finish and eliminated sharp edges or burrs. These finishing steps were essential to ensure that the collet fits accurately, grips workpieces securely, and functions smoothly in practical applications.

Conclusion

The collet manufacturing project was successfully completed using a universal lathe machine. All required machining operations turning, taper turning, centring, drilling, boring, and threading were performed accurately on a die steel workpiece. The final component met the desired dimensions and functional requirements.

This project enhanced practical understanding of lathe operations, tool handling, material properties, and precision machining. The successful manufacturing of the collet demonstrates the importance of proper material selection, accurate machining, and correct operation sequencing in mechanical manufacturing processes.